

Simon Xu

US Citizen | simondxu@gmail.com | Open to Relocation (Nationwide) | 626-621-7720
www.simondxu.com | linkedin.com/in/simondxu | github.com/simondxu

EXPERIENCE

Highmark Inc.

Aug 2024 - Present

Software Engineer (AI/ML team)

Remote, USA

- Shipped a real-time AI call-center training platform (**React/FastAPI/WebSockets**) where an LLM plays trainer-configured customer personas over the phone, replacing \$1M+/year of consultant-led training.
- Designed the rubric-based **LLM-as-judge** that scores each completed roleplay, tuning it against coach-scored transcripts until its scores tracked the coaches' before trainees saw any feedback.
- Built an enterprise **AI-agent lifecycle control plane** governing versioned releases from staging to production via a centralized registry, durable evaluation runs, policy-gated GitLab promotions, and cross-environment observability.
- Architected a **multi-agent evaluation harness** combining specialized reviewers, deterministic checks, and owner-defined and hidden tests to catch quality, safety, latency, and cost regressions before promotion.
- Created a team-shared, access-controlled knowledge workspace on **Vertex AI Search**, implementing a **RAG pipeline** with metadata-filtered retrieval, configurable chunking, and citation-grounded answers over user-uploaded documents.

Highmark Inc.

May 2024 – Aug 2024

Software Engineer Intern (AI/ML team)

Remote, USA

- Developed a **Retrieval-Augmented Generation** system using Flask, integrating **embedding models** and Gemini to answer employee queries on internal policies with cited sources; surveyed users reported 70% less search time.
- Engineered the internal **LLM API gateway** exposing a single Gemini endpoint to engineering teams company-wide, with schema-validated JSON output, per-consumer cost attribution, and circuit-breaking on provider errors.
- Built and scaled an internal **multimodal** AI assistant on Vertex AI (Gemini), enabling employees to process text, PDFs, images, and voice through a **React** and **FastAPI** stack; serves 30K users and processes 6M+ prompts annually.

PROJECTS

Proxy Loop: LLM Post-Training & Durable Agent Platform

Jul 2026 – Present

- Fine-tuned **Qwen3-8B** via **SFT + 4-bit QLoRA** on 1.5K provenance-tracked agent trajectories synthesized and filtered by a Claude Sonnet teacher; lifted held-out task completion 58%→67% and cut false completion 6%→2%.
- Architected a **Fast/Slow agent harness** separating low-latency voice interaction from long-horizon planning and tool use, with shared state, retries, approval gates, and completion checks across phone, browser, and email.
- Implemented durable long-running agent workflows with **Temporal + PostgreSQL**, persisting approval waits and scheduled follow-ups while using idempotent retries and callback deduplication to recover across worker restarts.

Conductor: Distributed Agent Execution Runtime (Go)

Apr 2026 – Jun 2026

- Engineered a **Go-based distributed agent runtime** with event-sourced recovery and idempotent tool execution, allowing long-running LLM workflows to resume after crashes without replaying completed side effects.
- Built a **Kubernetes**-native execution layer with PostgreSQL leases and fencing tokens to coordinate distributed workers, handling pod failures, graceful shutdowns, and rolling updates while preserving single-writer execution.
- Added end-to-end **OpenTelemetry + Prometheus observability** across model and tool calls, validating throughput, p95/p99 latency, and failure recovery through chaos and load testing.

EDUCATION

Georgia Institute of Technology — M.S. Computational Science & Engineering

Aug 2022 - Dec 2024

University of California - Santa Barbara — B.A. Applied Economics

Oct 2019 - Mar 2021

SKILLS

Languages	Python, TypeScript, JavaScript, Go, SQL, Agentic AI Coding (Claude Code, Codex, Cursor)
AI / LLM	RAG, agent harness engineering, LLM-as-Judge, agent evals, Fine-tuning, Embedding Models
Frameworks	PyTorch, Hugging Face Transformers, PEFT, vLLM, LangChain, Temporal, MCP, FastAPI, React
Data / Infra	PostgreSQL/pgvector, Redis, Docker, Kubernetes, CI/CD, AWS, GCP, OpenTelemetry